



Location Services

API User Document

Version 2.0

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Audience

This API is intended for web and mobile platforms.

Geography

Worldwide.

Background Technology

This API is a [REST](#)-based web service.

Response Format

This TWC API returns JSON formatted responses.

Location Services API

The Location Services API will query locations based on location name or geocode. The Location Services API returns information based on the user's search string entry.

Translations

This TWC API handles the translation of phrases. However, when formatting a request URL a valid language must be passed along (see the language code table for the supported codes).

Data Lifetime - Caching

Standard HTTP Cache-Control headers are used to define caching length. The TTL value is provided in the HTTP Header as an absolute time value using the "Expires" parameter, for example: "Expires: Fri, 12 Jul 2013 12:00:00 GMT"

Expiration

The response provides a data element expire_time_gmt. The value in this data element should be used to expire and remove a record from your system.

URL Construction

Please refer to the [TWC API Common Usage & Style Guide](#) for a reference and tutorial on URL construction and common terminology

Overview

The Location Services API provides the ability to lookup a location name or geocode (latitude and longitude) to retrieve a set of locations matching the request. The Location Service supports two decimal place precision for both latitude and longitude.

URL Format

Atomic API URL Examples: Your content licensing agreement with TWC determines the number of days returned in the API response and is constrained by the API Key that is provided to your company.	
Search by City Name: Returns information for search string ="atlanta".	Aggregate Product Name
https://api.weather.com/v2/location?address=atlanta&language=en-US&format=json&apiKey=yourApiKey	v2loc
Search by Geocode: Returns information for search string ="33.74,-84.39".	Aggregate Product Name
https://api.weather.com/v2/location?geocode=33.74,-84.39&language=en-US&format=json&apiKey=yourApiKey	v2loc
Search by Postal Code: Returns information for search string = "30339"	Aggregate Product Name
https://api.weather.com/v2/location?postalCode=30339&countryCode=US&language=en-US&format=json&apiKey=yourApiKey	v2loc

Query Parameters

The following represent query parameters for this API.

Parameter	Name	Description	Example
API Key	apiKey	The "apiKey" is used for access using this api. This key is provided to the customer.	https://api.weather.com/v2/location?geocode=33.74,-84.39&language=en-US&format=json&apiKey=yourApiKey
Language	language	The "language" parameter drives the language of the response. Please see the language code table below for valid values.	https://api.weather.com/v2/location?geocode=33.74,-84.39&language=en-US&format=json&apiKey=yourApiKey
Format	format	Only supported response format is: json	https://api.weather.com/v2/location?geocode=33.74,-84.39&language=en-US&format=json&apiKey=yourApiKey
Address	address	The city name to search.	https://api.weather.com/v2/location?address=atlanta&language=en-US&format=json&apiKey=yourApiKey
Geocode	geocode	The latitude and longitude.	https://api.weather.com/v2/location?geocode=33.74,-84.39&language=en-US&format=json&apiKey=yourApiKey NOTE: Only use a period (.) to indicate precision for a latitude or longitude (comma is not supported as a precision indicator).
Postal Code	postalCode	The postal code to search, must be used in combination with the countryCode parameter.	https://api.weather.com/v2/location?postalCode=30301&countryCode=US&language=en-US&format=json&apiKey=yourApiKey
Country Code	countryCode	The country code in which the postal code resides, must be used in combination with the postalCode parameter.	https://api.weather.com/v2/location?postalCode=30301&countryCode=US&language=en-US&format=json&apiKey=yourApiKey

Cross-Origin Resource Sharing

All TWC APIs support Cross-Origin Resource Sharing or CORS. For additional information on this communications mechanism, please refer to the following documents:

W3 CORS Specification

- <http://www.w3.org/TR/cors/>

CORS tutorial

- <http://www.html5rocks.com/en/tutorials/cors/>

Error Handling

The response on error is always the same, allowing multiple error types to be returned within one response:

JSON Example:

```
{
  metadata: {
    transaction_id: "1411496413365:-1880721071"
  },
  success: false,
  errors: [
    {
      error: {
        code: "NDF-0001",
        message: "There was no data found for your <product> query."
      }
    }
  ]
}
```

HTTP error responses common to all requests are as follows:

Error Code	Description
200	OK. The request has succeeded.
204	No Data Found for specific query. The 204 status code will have an empty response body.
400	Bad request. The request could not be understood by the server due to malformed syntax. The API will reject the request if any invalid parameters are supplied.
401	Unauthorized. The request requires authentication.
403	Forbidden. The server understood the request but is refusing to fulfill it.
404	Not found. Any required parameter that is not present in the API request will result in a MissingParameterException being thrown with a status of 404 returned.
500	Internal server error. The server encountered an unexpected condition which prevented it from fulfilling the request.
502	Bad gateway.
503	Service unavailable.
504	Gateway timeout.

Response Field Maintenance

TWC strives to minimize the impact of changes in our weather content to your applications. TWC will not remove, rename or change the data type (int, string) of any data fields in the API response. However, TWC may add new data fields without notice.

Response Field Data Elements & Descriptions

Outbound JSON/XML	Description	Type	Optional	Sample
Metadata				
language	Requested language that is passed as a parameter within the URL	string	Y	en-US
transaction_id	API Transaction ID	string	Y	
version	API version	string	Y	v2
format	Format of the response	string	Y	json
latitude	Latitude of requested location that is passed as a parameter within the URL. Populated only if the API is called via geocode (Latitude/Longitude)	double	Y	33.75
longitude	Longitude of requested location that is passed as a parameter within the API url. Populated only if the API is called via geocode (Latitude/Longitude)	double	Y	-84.39
address	Address of requested location that is passed as a parameter within the URL. Populated only if the API is called via Address	string	Y	Atlanta
status_code	Status code	integer	Y	200
	The following fields may repeat several times			
latitude	The Location's Latitude	double	Y	33.75
longitude	The Location's Longitude	double	Y	-84.39
address	The street address	string	Y	100 Main Street
locality	The city or place name	string	Y	Atlanta
admin_district	The internationalized identifier for 'state' or geopolitical area - level 1 administrative division.	string	Y	GA, Georgia
postal_code	The postal code	string	Y	30308
country	Full name of location Country	string	Y	United States
country_code	The country code abbreviation.	string	Y	US

Formatted Response Sample

JSON Example By Geocode	JSON Example By Address
https://api.weather.com/v2/location?geocode=33.74,-84.39&language=en-US&format=json&apiKey=yourApiKey <pre>{ "metadata": { "version": "v2", "transaction_id": "1461167809540:-538469089", "generated_time": 1461167809, "total_cache_time_secs": 86400, "status_code": 200, "language": "en-US", "latitude": 33.74, "longitude": -84.39, "format": "json" }, "addresses": [{ "latitude": 33.74, "longitude": -84.39, "locality": "Atlanta", "admin_district": "GA", "postal_code": "30312", "country": "United States", "country_code": "US" }] }</pre>	https://api.weather.com/v2/location?address=Atlanta&language=en-US&format=json&apiKey=yourApiKey <pre>{ "metadata": { "version": "v2", "transaction_id": "1461241815014:993484287", "generated_time": 1461241815, "total_cache_time_secs": 86400, "status_code": 200, "language": "en-US", "address": "Atlanta", "format": "json" }, "addresses": [{ "latitude": 33.74831008911133, "longitude": -84.39111328125, "locality": "Atlanta", "admin_district": "GA", "country": "United States" }, { "latitude": 33.115299224853516, "longitude": -94.1681137084961, "locality": "Atlanta", "admin_district": "TX", "country": "United States" }, { "latitude": 40.26369094848633, "longitude": -89.23509979248047, "locality": "Atlanta", "admin_district": "IL", "country": "United States" }] }</pre>